Figure 2.4 **Generic decision tree for identification of appropriate tier to estimate changes in carbon stocks in mineral soils by land-use category.**

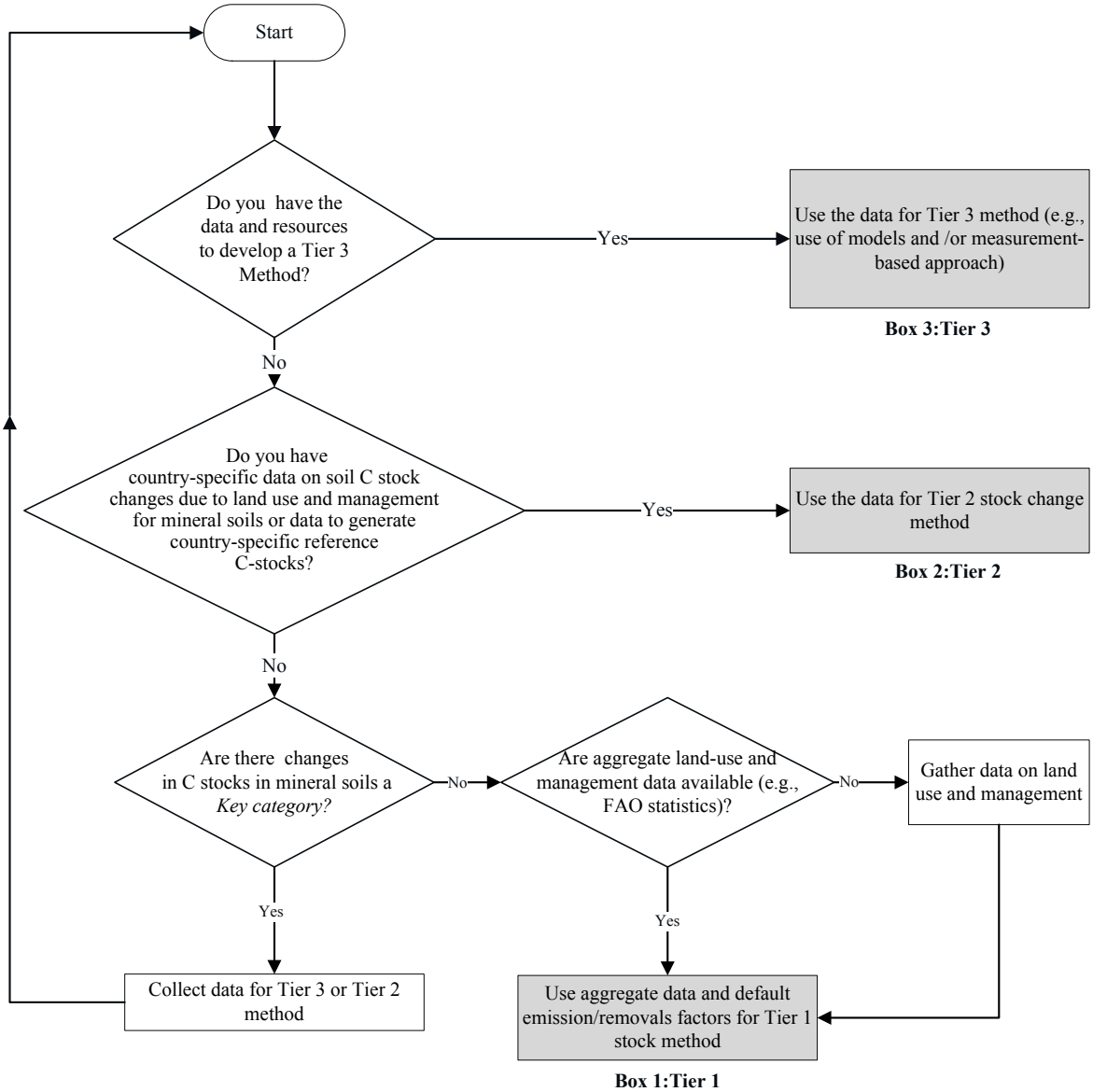
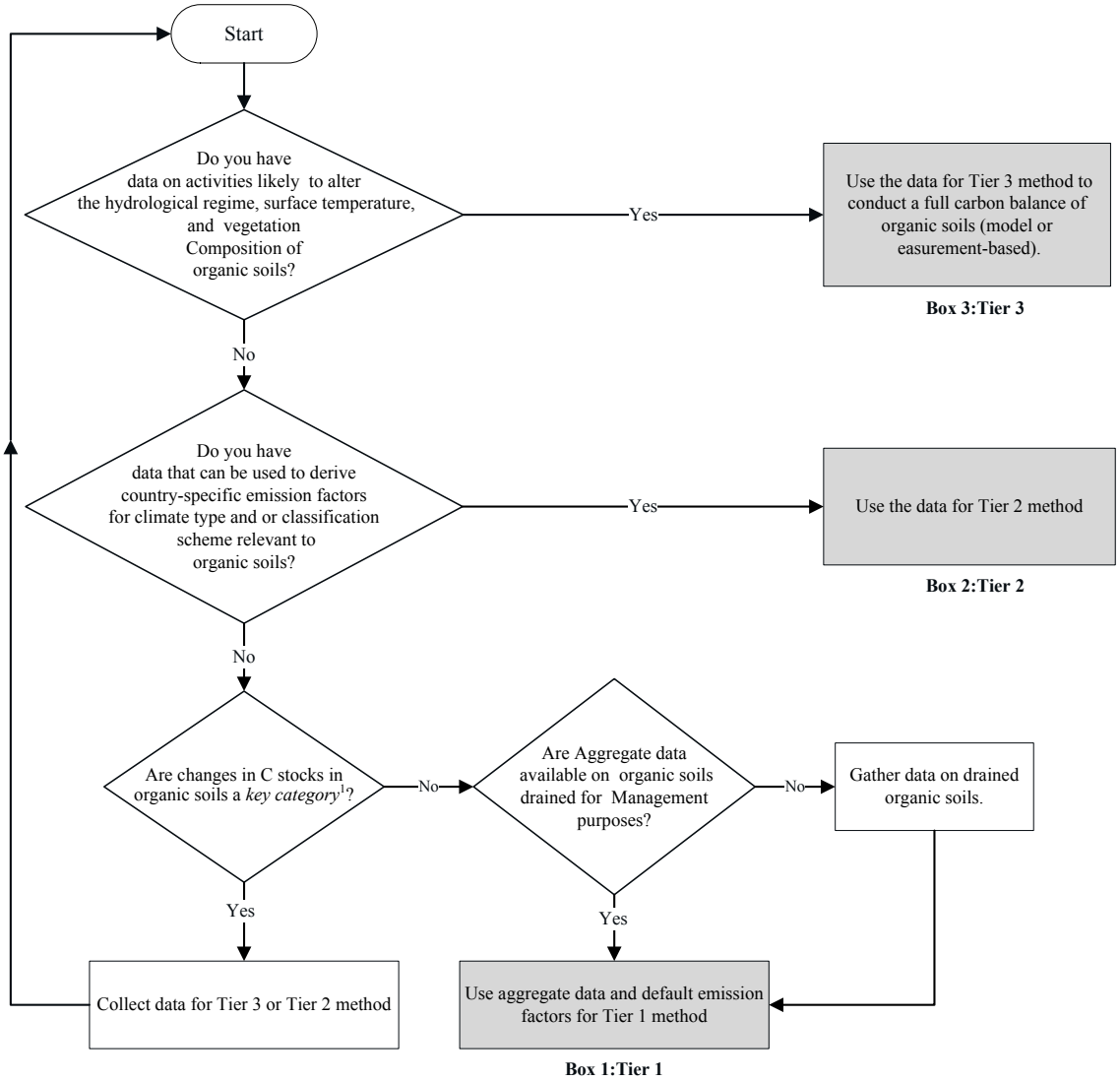


Figure 2.5 Generic decision tree for identification of appropriate tier to estimate changes in carbon stocks in organic soils by land-use category



Tier 1 Default Method

Mineral soils

For mineral soils, the stock change factor method is based on changes in soil C stocks ( ) over a finite period of time of 20 years (Equation 2.25). The change in organic C stock in mineral soil ( ) is computed by calculating the organic C stock remaining after a management change relative to the organic C stock in a reference condition and summing this change over all climate zones, soil types and management practices included in the inventory. The soil organic C stock present under the reference condition for the Tier 1 method is defined as that in non-degraded, unimproved lands under native vegetation (Table 2.3). The following assumptions are made:

(i) Over time, soil organic C stock reaches a spatially-averaged, stable value specific to the soil, climate, land-use and management practices; and

(ii) Soil organic C stock change during the transition to a new equilibrium SOC occurs in a linear fashion over a period of 20 years.

Assumption (i), that under a given set of climate and management conditions soils tend towards an equilibrium organic C stock, is widely accepted. Although, soil organic C stock changes in response to management changes may often be best described by a curvilinear function, assumption (ii) greatly simplifies the Tier 1 methodology and provides a good approximation over a multi-year inventory period, where changes in management and land-use conversions are occurring throughout the inventory period.

TABLE 2.3 (UPDATED)
DEFAULT REFERENCE CONDITION SOIL ORGANIC CARBON STOCKS (SOC_{REF}) FOR MINERAL SOILS (TONNES C HA^{-1} IN 0-30 CM DEPTH) [1, 2]

IPCC Climate Zone [5]	IPCC soil class [6]		
	High activity clay soils (HAC) [7]	**Low activity clay soils (LAC) [8]**	**Sandy soils (SAN) [9]**
Polar Moist/Dry (Px - undiff)[13]	59 ± 41% (24)	NA	27 ± 67% (18)
Boreal Moist/Dry (Bx - undiff)[13]	63 ± 18% (35)	NA	10 ± 90% [4]
Cool temperate dry (C2)	43 ± 8% (177)	33 ± 90% [3]	13 ± 33% (10)
Cool temperate moist (C1)	81 ± 5% (334)	76 ± 51% (6)	51 ± 13% (126)
Warm temperate dry (W2)	24 ± 5% (781)	19 ± 16% (41)	10 ± 5% (338)
Warm temperate moist (W1)	64 ± 5% (489)	55 ± 8% (183)	36 ± 23% (39)
Tropical dry (T4)	21 ± 5% (554)	19 ± 10% (135)	9 ± 9% (164)
Tropical moist (T3)	40 ± 7% (226)	38 ± 5% (326)	27 ± 12% (76)
Tropical wet (T2)	60 ± 8% (137)	52 ± 6% (271)	46 ± 20% (43)
Tropical montane (T1)	51 ± 10% (114)	44 ± 11% (84)	52 ± 34% (11)
	Spodic soils (POD) [10]	**Volcanic soils (VOL) [11]**	**Wetland soils (WET) [12]**
Polar Moist/Dry (Px - undiff)[13]	NO	NA	NA
Boreal Moist/Dry (Bx - undiff)[13]	117 ± 90% [3]	20 ± 90% [4]	116 ± 65% (6)
Cool temperate dry (C2)	NO	20 ± 90% [4]	87 ± 90% [3]
Cool temperate moist (C1)	128 ± 14% (45)	136 ± 14% (28)	128 ± 13% (42)
Warm temperate dry (W2)	NO	84 ± 65% (10)	74 ± 17% (49)
Warm temperate moist (W1)	143 ± 30% (9)	138 ± 12% (42)	135 ± 28% (28)
Tropical dry (T4)	NA	50 ± 90% [4]	22 ± 17% (32)
Tropical moist (T3)	NA	70 ± 90% [4]	68 ± 17% (55)
Tropical wet (T2)	NA	77 ± 27% (14)	49 ± 19% (33)
Tropical montane (T1)	NA	96 ± 31% (10)	82 ± 50% (12)

Note: Data are derived from Batjes (2010) and Batjes (2011) unless otherwise noted through the use of superscripts.

[1] NA denotes that soil categories the soil category may occur in a climate zone, but no data was available. NO denotes that the soil type does not normally occur within a climate zone. [2] All values are presented in the format of the mean for the soil by climate combination ± the 95% confidence limit expressed as a percentage of the mean (that is ± 1.96 * standard error /mean *100). Values in parentheses are the number of soils included in the derivation of mean and standard error values for each combination of soil and climate types. [3] Indicates where no data were available from Batjes (2011) but values were derived for the *2006 IPCC Guidelines for National Greenhouse Gas Inventories* and have been used in the table. No values of n were available. A nominal error estimate of ±90% of the mean was assigned as per the *2006 IPCC Guidelines*. [4] Indicates where no data were available either from Batjes (2011) or in the *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Mean values present the default values used in the 1996 IPCC Guidelines. No values of n were available. A nominal error estimate of ±90% of the mean was assigned as per the *2006 IPCC Guidelines*. [5] Climate classes are defined according to (IPCC 2006, p. 3.39) using elevation, mean annual temperature, mean annual precipitation, mean annual precipitation to potential evapotranspiration ratio and frost occurrence. [6] Soil classes are inferred from the FAO-1990/WRB-2006 classification in accordance with IPCC (2006, p. 3.40 - 3.41). [7] Soils with high activity clay (HAC) minerals are lightly to moderately weathered soils dominated by 2:1 silicate clay minerals (in the World Reference Base for Soil Resources (WRB) classification: Leptosols, Vertisols, Kastanozems, Chernozems, Phaeozems, Luvisols, Alisols, Albeluvisols, Solonetz, Calcisols, Gypsisols, Umbrisols, Cambisols, Regosols; in USDA classification: Mollisols, Vertisols, high-base status Alfisols, Aridisols, Inceptisols). [8] Soils with low activity clay (LAC) minerals are highly weathered soils, dominated by 1:1 clay minerals and amorphous iron and aluminium oxides (in WRB classification: Acrisols, Lixisols, Nitisols, Ferralsols, Durisols; in USDA classification: Ultisols, Oxisols, acidic Alfisols). [9] Soils (regardless of taxonomic classification) having > 70% sand and < 8% clay (in WRB classification: Arenosols; in USDA classification: Psamments). [10] Soils exhibiting strong podzolization (in WRB classification includes Podzols; in USDA classification Spodosols). [11] Soils derived from volcanic ash with allophanic mineralogy (in WRB classification Andosols; in USDA classification Andisols). [12] Soils with restricted drainage leading to periodic flooding and anaerobic conditions (in WRB classification Gleysols; in USDA classification Aquic suborders). [13] The Boreal dry and Boreal moist zones and the Polar dry and Polar moist zones were not differentiated. Results presented represent the SOC_{30} stocks for the undifferentiated (undiff.) Boreal (Bx) and Polar (Px) classes.

(higher tiers) which includes the dynamics of dead organic matter and improves the estimates of direct and post-fire emissions.

For Forest Land converted to other land uses, organic matter burnt is derived from both newly felled vegetation and existing dead organic matter, and CO_2 emissions should be reported. In this situation, estimates of total fuel consumed (Table 2.6) can be used to estimate emissions of CO_2 and non- greenhouse gases using Equation 2.27. Care must be taken, however, to ensure that dead organic matter carbon losses during the land-use conversion are not double counted in Equations 2.27 (as losses from burning) and Equation 2.23 (as losses from decay).

A generic methodology to estimate the emissions of individual greenhouse gases for any type of fire is summarised in Equation 2.27.

EQUATION 2.27
ESTIMATION OF GREENHOUSE GAS EMISSIONS FROM FIRE

$$L_{fire} \quad A \quad M_B \quad C_f \quad G_{ef} \quad 10^{3}$$

Where:

L_{fire} = amount of greenhouse gas emissions from fire, tonnes of each GHG e.g., CH_4, N_2O, etc.

A = area burnt, ha

M_B = mass of fuel available for combustion, tonnes ha^{-1}. This includes biomass, ground litter and dead wood. When Tier 1 methods are used then litter and dead wood pools are assumed zero, except where there is a land-use change (see Section 2.3.2.2).

C_f = combustion factor, dimensionless (default values in Table 2.6)

G_{ef} = emission factor, g kg^{-1} dry matter burnt (default values in Table 2.5)

Note. Where data for M_B and C_f are not available, a default value for the amount of fuel actually burnt (the product of M_B and C_f) can be used (Table 2.4) under Tier 1 methodology.

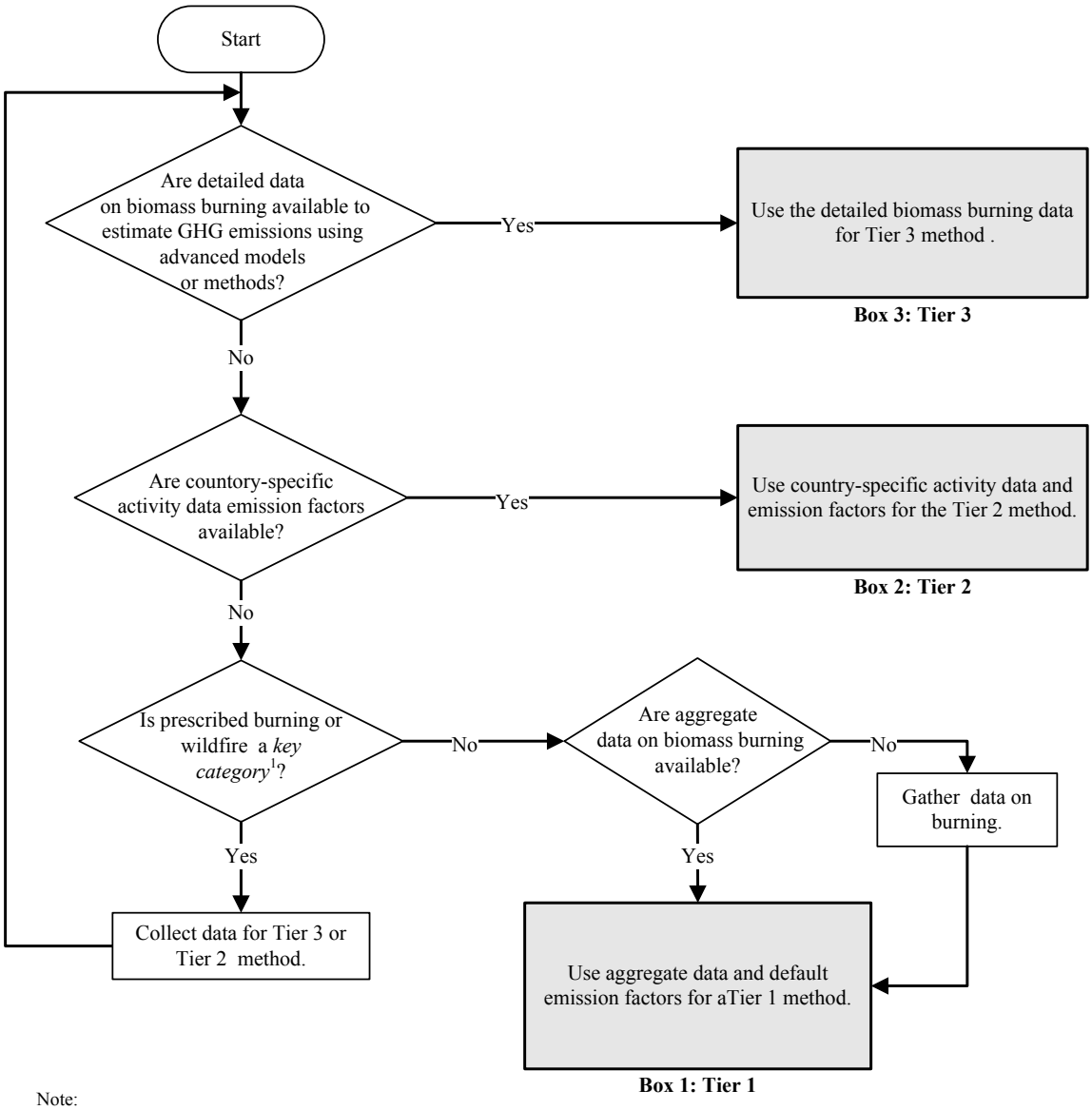
For CO_2 emissions, Equation 2.27 relates to Equation 2.14, which estimates the annual amount of live biomass loss from any type of disturbance.

The amount of fuel that can be burnt is given by the area burnt and the density of fuel present on that area. The fuel density can include biomass, dead wood and litter, which vary as a function of the type, age and condition of the vegetation. The type of fire also affects the amount of fuel available for combustion. For example, fuel available for low-intensity ground fires in forests will be largely restricted to litter and dead organic matter on the surface, while a higher-

The combustion factor is a measure of the proportion of the fuel that is actually combusted, which varies as a function of the size and architecture of the fuel load (i.e., a smaller proportion of large, coarse fuel such as tree stems will be burnt compared to fine fuels, such as grass leaves), the moisture content of the fuel and the type of fire (i.e., intensity and rate of spread which is markedly affected by climatic variability and regional differences as reflected in Table 2.4). Finally, the emission factor gives the amount of a particular greenhouse gas emitted per unit of dry matter combusted, which can vary as a function of the carbon content of the biomass and the completeness of combustion. For species with high N concentrations, NO_x and N_2O emissions from fire can vary as a function of the N content of the fuel. A comprehensive review of emission factors was conducted by Andreae and Merlet (2001) and is summarised in Table 2.5.

Tier 2 methods employ the same general approach as Tier 1 but make use of more refined country-derived emission factors and/or more refined estimates of fuel densities and combustion factors than those provided in the default tables. Tier 3 methods are more comprehensive and include considerations of the dynamics of fuels (biomass and dead organic matter).

Figure 2.6 Generic decision tree for identification of appropriate tier to estimate greenhouse gas emissions from fire in a land-use category



Note:
1. See Volume 1 Chapter 4, Methodological Choice and Identification of Key Categories (noting Section 4.1.2 on limited resources), for discussion of *key categories* and use of decision trees.

TABLE 2.4 (UPDATED)
FUEL (DEAD ORGANIC MATTER PLUS LIVE BIOMASS) BIOMASS CONSUMPTION VALUES (TONNES DRY MATTER HA^{-1}) FOR FIRES IN A RANGE OF VEGETATION TYPES

(To be used in Equation 2.27, to estimate the SURGXFWRI WXDQWLW HOHPHQW RI EXUQHG PDWHULDO

Vegetation type	Subcategory	Mean	SE	References
Primary tropical forest (slash and burn)	Primary tropical forest	83.9	25.8	7, 15, 66, 3, 16, 17, 45
	Primary open tropical forest	163.6	52.1	21,
	Primary tropical moist forest	160.4	11.8	37, 73
	Primary tropical dry forest	-	-	66
All primary tropical forests		**119.6**	**50.7**	
Secondary tropical forest (slash and burn)	Young secondary tropical forest (3-5 yrs)	8.1	-	61
	Intermediate secondary tropical forest (6-10 yrs)	41.1	27.4	61, 35
	Advanced secondary tropical forest (14-17 yrs)	46.4	8.0	61, 73
All secondary tropical forests		**42.2**	**23.6**	66, 30
All Tertiary tropical forest		**54.1**	**-**	66, 30
Boreal forest	Wildfire (general)	52.8	48.4	2, 33, 66
	Crown fire	25.1	7.9	11, 43, 66, 41, 63, 64
	Surface fire	21.6	25.1	43, 69, 66, 63, 64, 1
	Post logging slash burn	69.6	44.8	49, 40, 66, 18
	Land clearing fire	87.5	35.0	10, 67
All boreal forest		**41.0**	**36.5**	43, 45, 69, 47
Eucalypt forests	Wildfire	53.0	53.6	66, 32, 9
	Prescribed fire (surface)	16.0	13.7	66, 72, 54, 60, 9
	Post logging slash burn	168.4	168.8	25, 58, 46
	Felled, wood removed, and burned (land-clearing fire)	132.6	-	62, 9
All Eucalypt forests		**69.4**	**100.**	
Other temperate forests	Wildfire	19.8	6.3	32, 66
	Post logging slash burn	77.5	65.0	55, 19, 14, 27, 66
	Felled and burned (land-clearing fire)	48.4	62.7	53, 24, 71
		50.4	**53.7**	43, 56
Shrublands	Shrubland (general)	26.7	4.2	43
	Calluna heath	11.5	4.3	26, 39
	Sagebrush	5.7	3.8	66
	Fynbos	12.9	0.1	70, 66
All Shrublands		**14.3**	**9.0**	

TABLE 2.4 (UPDATED) (CONTINUED)
FUEL (DEAD ORGANIC MATTER PLUS LIVE BIOMASS) BIOMASS CONSUMPTION VALUES (TONNES DRY MATTER HA^{-1}) FOR FIRES IN A RANGE OF VEGETATION TYPES

(To be used in Equation 2.27, WRHVWLPDWHWKHSURGXFWRIWKHXQEXUQHGIXHOPDVVXVHGRXW

Vegetation type	Subcategory	Mea	SE	References
Savanna woodlands (early dry season burns)*	Savanna woodland	2.5	-	28
	Savanna parkland	2.7	-	57
All savanna woodlands (early dry season burns)		**2.6**	**0.1**	
Savanna woodlands (mid/late dry season burns)*	Savanna woodland	3.3	-	57
	Savanna parkland	4.0	1.1	57, 6, 51
	Tropical savanna	6	1.8	52, 73
	Other savanna woodlands	5.3	1.7	59, 57, 31
All savanna woodlands (mid/late dry season burns)*		**4.6**	**1.5**	
Savanna Grasslands/ Pastures (early dry season burns)*	Tropical/sub-tropical grassland	2.1	-	28
	Grassland	-	-	48
All savanna grasslands (early dry season burns)*		**2.1**	**-**	
Savanna Grasslands/ Pastures (mid/late dry season burns)*	Tropical/sub-tropical grassland	5.2	1.7	9, 73, 12, 57
	Grassland	4.1	3.1	43, 9
	Tropical pasture~	23.7	11.8	4, 23, 38, 66
	Savanna	7.0	2.7	42, 50, 6, 45, 13, 65
All savanna grasslands (mid/late dry season burns)*		**10.0**	**10.1**	
Other vegetation types	Peatland	41	1.4	68, 33
	Tundra	10	-	33
Agricultural residues (post-harvest field burning)	$M_B = AGR_{(T)} \times Frac_{Brunt(T)}$			See Equation 11.6 in Chapter 11, Volume 4 for $AGR_{(T)}$ calculation

* Surface layer combustion only

~Derived from slashed tropical forest (includes unburned woody material)

[a] For sugarcane, data refer to burning before harvest of the crop.

[b] Expert assessment by authors.

TABLE 2.5
EMISSION FACTORS (g kg^{-1} DRY MATTER BURNT) FOR VARIOUS TYPES OF BURNING. VALUES ARE MEANS ± SD AND ARE BASED ON THE COMPREHENSIVE REVIEW BY ANDREAE AND MERLET (2001)
7REHXVHGDVTXDQWLW (see Equation 2.27)

Category	CO_2	CO	CH_4	N_2O	NO_X
Savanna and grassland	1613 ± 95	65 ± 20	2.3 ± 0.9	0.21 ± 0.10	3.9 ± 2.4
Agricultural residues	1515 ± 177	92 ± 84	2.7	0.07	2.5 ± 1.0
Tropical forest	1580 ± 90	104 ± 20	6.8 ± 2.0	0.20	1.6 ± 0.7
Extra tropical forest	1569 ± 131	107 ± 37	4.7 ± 1.9	0.26 ±0.07	3.0 ± 1.4
Biofuel burning	1550 ± 95	78 ± 31	6.1 ± 2.2	0.06	1.1 ± 0.6

Note: For combustion of non-woody biomass in Grassland and Cropland, CO_2 emissions do not need to be estimated and reported, because it is assumed that annual CO_2 removals (through growth) and emissions (whether by decay or fire) by biomass are in balance (see earlier discussion on synchrony in Section 2.4.

TABLE 2.6 (UPDATED)
COMBUSTION FACTOR VALUES (PROPORTION OF PREFIRE FUEL BIOMASS CONSUMED) FOR FIRES IN A RANGE OF VEGETATION TYPES
9DOXHV &RQFRXQQHQGDUHW &RHIIXVHG &I (Cf in Equation 2.27)

Vegetation type	Subcategory	Mean	SD	References
Primary tropical forest (slash and burn)	Primary tropical forest	0.32	0.12	7, 8, 15, 56, 66, 3, 16, 53, 17, 45,
	Primary open tropical forest	0.45	0.09	21
	Primary tropical moist forest	0.50	0.03	37, 73
	Primary tropical dry forest	-	-	66
All primary tropical forests		**0.36**	**0.13**	
Secondary tropical forest (slash and burn)	Young secondary tropical forest (3-5 yrs)	0.46	-	61
	Intermediate secondary tropical forest (6-10 yrs)	0.67	0.21	61, 35
	Advanced secondary tropical forest (14-17 yrs)	0.50	0.10	61, 73
All secondary tropical forests		**0.55**	**0.06**	56, 66, 34, 30
All tertiary tropical forest		**0.59**	**-**	66, 30
Boreal forest	Wildfire (general)	0.40	0.06	33
	Crown fire	0.43	0.21	66, 41, 64, 63
	surface fire	0.15	0.08	64, 63
	Post logging slash burn	0.33	0.13	49, 40, 18
	Land clearing fire	0.59	-	67
All boreal forest		**0.34**	**0.17**	45, 47
Eucalyptus forests	Wildfire	-	-	
	Prescribed fire (surface)	0.61	0.11	72, 54, 60, 9
	Post logging slash burn	0.68	0.14	25, 58, 46
	Felled and burned (land-clearing fire)	0.49	-	62
All Eucalyptus forests		**0.63**	**0.13**	
Other temperate forests	Post logging slash burn	0.62	0.12	55, 19, 27, 14
	Felled and burned (land-clearing fire)	0.51	-	53, 24, 71
		0.45	**0.16**	53, 56

TABLE 2.6 (UPDATED) (CONTINUED)
COMBUSTION FACTOR VALUES (PROPORTION OF PREFIRE FUEL BIOMASS CONSUMED) FOR FIRES IN A RANGE OF VEGETATION TYPES
9DOXHV RI FRPEXVWLRQ IDFWRU IRU XVH LQ &RPSXWLQJ &f (Refer to Equation 2.27)

Vegetation type	Subcategory	Mean	SD	References
Shrublands	Shrubland (general)	0.95	-	44
	Calluna heath	0.71	0.30	26, 56, 39
	Fynbos	0.61	0.16	70, 44
All shrublands		**0.72**	**0.25**	
Savanna woodlands (early dry season burns)*	Savanna woodland	0.22	-	28
	Savanna parkland	0.73	-	57
	Other savanna woodlands	0.37	0.19	22, 29
All savanna woodlands (early dry season burns)		**0.40**	**0.22**	
Savanna woodlands (mid/late dry season burns)*	Savanna woodland	0.72	-	66, 57
	Savanna parkland	0.82	0.07	57, 6, 51
	Tropical savanna	0.73	0.04	52, 73, 66, 12
	Other savanna woodlands	0.68	0.19	22, 29, 44, 31, 57
All savanna woodlands (mid/late dry season burns)*		**0.74**	**0.14**	
Savanna Grasslands/ Pastures (early dry season burns)*	Tropical/sub-tropical grassland	0.74	-	28
	Grassland	-	-	48
All savanna grasslands (early dry season burns)*		**0.74**	**-**	
Savanna Grasslands/ Pastures (mid/late dry season burns)*	Tropical/sub-tropical grassland	0.92	0.11	44, 73, 66, 12, 57
	Tropical pasture~	0.35	0.21	4, 23, 38, 66
	Savanna	0.86	0.12	53, 5, 56, 42, 50, 6, 45, 13, 44, 65, 66
All savanna grasslands (mid/late dry season burns)*		**0.77**	**0.26**	
Other vegetation types	Peatland	0.50	-	20, 44
	Tropical Wetlands	0.70	-	44
Agricultural residues (Post-harvest field burning)	Wheat residues	0.90	-	see Note b
	Maize residues	0.80	-	see Note b
	Rice residues	0.80	-	see Note b
	Sugarcane a	0.80	-	see Note b
	Other Crops	0.85	-	see Note b

* Surface layer combustion only; ~ Derived from slashed tropical forest (includes unburned woody material); [a] For sugarcane, data refer to burning before harvest of the crop; [b] Expert assessment by authors.

2.6 INTER-ANNUAL VARIABILITY

In the AFOLU sector, the management of land is used as the best approximation of human influence and thus, estimates of emissions and removals on managed land are used as a proxy for anthropogenic emissions and removals on the basis that the preponderance of anthropogenic effects occurs on managed lands (see Vol. 4 Chapter 1). This allows for consistency, comparability, and transparency in estimation. Referred to as the Managed Land Proxy (MLP), this approach is currently recognised by the IPCC as the only universally applicable approach to estimating anthropogenic emissions and removals in the AFOLU sector (IPCC 2006, IPCC 2010). However, it is also recognised that the estimated emissions and removals on managed lands can represent a combination of both anthropogenic (direct and indirect) and natural effects (Vol. 4 Chapter 1 p1.5; IPCC 2010; see Fig. 2.6A).

Some of the emissions and removals from managed land are characterised by high interannual variability. Interannual variability (IAV) refers to the variability in the annual emissions and removals (E/R) estimates between years within a time series. In the AFOLU sector, the application of the MLP means that IAV can be caused by both anthropogenic and natural causes. The three main causes of IAV in GHG emissions and removals in the AFOLU sector are (1) natural disturbances (such as wildfires, insects, windthrow, and ice storms), which can cause large immediate and delayed emissions and kill trees; (2) climate variability (e.g. temperature, precipitation, and drought), which affects photosynthesis and respiration (Ciais *et al.* 2005; Aragão *et al.* 2018); and, (3) variation in the rate of human activities, including land use (such as forest harvesting), and land-use change (Stinson *et al.* 2011; Pilli *et al.* 2016; Kurz *et al.* 2018).

In some countries IAV in emissions from natural disturbances can be larger than the IAV of emissions caused by human activities such as 0 to 2016 time series of annual emission and removals due to natural disturbances is much larger than the IAV in the emissions and removals on the remaining managed forest land (Figure 2.6C). The NGHGIs for Portugal (Figure 6- (Portuguese Environmental Agency 2018) (Commonwealth of Australia 2018)) are two other examples of time series with high IAV. In some countries, the areas burned by wildfires can vary by two orders of magnitude between years (Stinson *et al.* 2011; Miller *et al.* 2012; Genet *et al.* 2018). In other countries, the majority of IAV may be due to human activities.

When the MLP is used and the IAV in emissions and removals due to natural disturbances is large, it is difficult to gain a quantitative understanding of the role of human activities compared to the impacts of natural effects. In such situations, disaggregating[9] MLP emissions and removals into those that are considered to result from human activities and those understood to result from natural effects may provide increased understanding of the emissions and removals that are due to human activities such as, land use (including harvesting) and land-use change. In this way, disaggregation can contribute to improved understanding of the trends in emission and removals due to human activities and mitigation actions that are taken to reduce anthropogenic emissions and preserve and enhance carbon stocks.

Disaggregating emissions and removals according to anthropogenic and natural effects has been recognised as a scientific challenge (Canadell *et al.* 2007; Vetter *et al.* 2008; IPCC 2010; Kurz 2010; Smith 2010; Brando *et al.* 2014; Henttonen *et al.* 2017). It is not yet possible to fully and accurately separate emissions and removals associated with human activity from those associated with natural effects. The last IPCC Expert Meeting Report on this topic encouraged further development of scientific methods (IPCC 2010).

Recognizing that some but not all countries may choose to address emissions and removals from natural disturbances on managed land outside the inventory process, this guidance is provided as an option that may be used by countries that choose to disaggregate their reported MLP emissions and removals (i.e. all emissions and removals on managed land) into those that are considered to result from human activities and those that are considered to result from natural disturbances. These supplementary approaches may be of interest to countries with AFOLU sector emissions where IAV due to natural effects is large. The section first addresses definitional issues, followed by a description of whether or not different methodological approaches used to estimate C stock changes quantify the IAV of emissions and removals. A generic approach to report on disaggregation of the contribution of natural disturbances in reporting on total emissions and removals on managed lands is then provided, along with country-specific examples of methodological approaches to disaggregating anthropogenic effects and natural disturbances on managed lands.

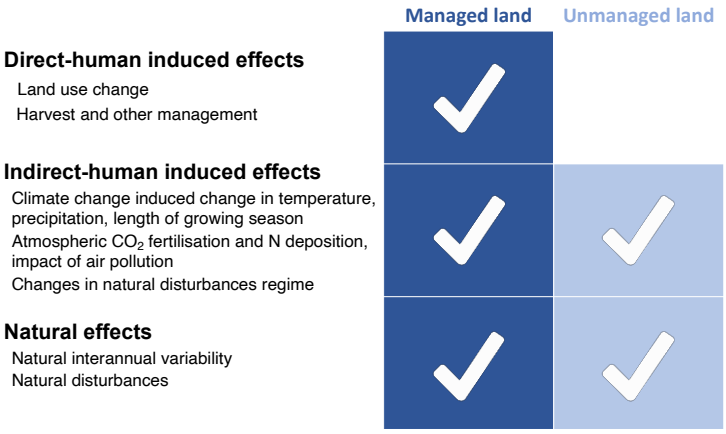
[9] Disaggregating means that an estimate is separated into its component parts.

2.6.1 Definitional issues

2.6.1.1 DIRECT AND INDIRECT HUMAN EFFECTS, AND NATURAL EFFECTS

Anthropogenic (i.e., direct and indirect human) effects and natural effects are described in Vol. 4 Chapter 1. Figure 2.6a summarizes the main factors that cause these effects and their occurrences in managed and unmanaged lands. The specific effects included in estimates reported in NGHGIs depend on the estimation method and data used, which differ in approach and complexity among countries (see Table 2.6c). Describing how the various effects are reflected in the estimates of emissions and removals, based on the estimation method and data used, increases the transparency of the NGHGI and its understanding by the scientific and policy communities (Grassi *et al.* 2018, section 2.6.2). Useful information may include definition and spatial maps of managed land, information on areas of forest being harvested and those subject to other management, and information on the main determinants of the GHG fluxes (e.g., forest age structure, harvested volumes, harvest cycle).

Figure 2.6a: Conceptual illustration of how various anthropogenic (direct and indirect) and natural factors affect land-related GHG emissions and removals in managed and unmanaged lands (Source: Grassi *et al.* (2018)).



Direct human-induced effects of any management activity on emissions or removals, by definition, only occur on managed lands. Indirect human-induced effects (i.e., the second order impacts of human activities on emissions or removals mediated through environmental change) and natural effects can occur on both unmanaged and managed lands. The (Vol. 4, Ch. 1). The natural effects (Vol. 4, Ch. 1), provided that there are no trends in disturbance rates, such as increased annual area burned as a result of climate change. Nonetheless, their IAV in emissions and removals can have an important impact on annual NGHGIs. Depending on the estimation method and data used, GHG estimates for managed land may capture all or only some of this IAV (see Section 2.6.2).

The IPCC describes the MLP as a method to approximate estimates of anthropogenic emissions and removals, but this proxy estimate also contains emissions and removals resulting from natural disturbances (IPCC 2006; IPCC 2010). This section introduces an approach that countries can apply on a voluntary basis within the MLP in order to indicate those emissions and removals considered to result from human activity, and those that are understood to result from natural disturbances. This is achieved by disaggregating the estimated emissions and removals due to natural disturbances (ND E/R) within the estimated total MLP emissions and removals. This remaining aggregate of emissions and removals associated with human activity might still include some effects of IAV of natural disturbances and other natural effects on anthropogenic emissions and removals.

2.6.1.2 NATURAL DISTURBANCES

Disturbances, in particular wildfires, can contribute to large IAV in emissions. The number, frequency and intensity of fire events are strongly controlled by climate and weather, fuels, ignition sources, and human activities. High temperatures, past levels of fire suppression, and persistent drought events are key drivers of forest fires, for

instance in the Western US (Westerling 2016), in the Amazon region (Morton *et al.* 2013) or in Indonesia (Schimel *et al.* 2015). However, land use and land-use change such as deforestation and peatland drainage can influence the likelihood and impacts of fire (Page & Hooijer 2016). In the Brazilian Cerrado, severe drought events explain the loss of almost 30 percent of aboveground woody biomass (de Miranda *et al.* 2014). Other natural disturbances with large IAV include storm damage (Yamashita *et al.* 2002; Lindner *et al.* 2010). Insects tend to follow outbreak cycles, thus causing more long-term trends that contribute to interdecadal rather than interannual variations (Kurz *et al.* 2008; Hicke *et al.* 2012). However, like IAV, the inter-decadal variability can also make it difficult to identify trends in emissions and removals that result from human activities.

Definition of natural disturbances

Natural disturbances in the context of the AFOLU sector are non-anthropogenic events or non-anthropogenic circumstances that cause significant emissions and are beyond the control of, and not materially influenced by a country. These include wildfires, insect and disease infestations, extreme weather events and/or geological disturbances, beyond the control of, and not materially influenced by a country. Natural disturbances exclude human activities such as harvesting, prescribed burning and fires associated with activities such as slash and burn.[10]

Non-anthropogenic events refer to non-human induced events (e.g. fire initiated by lightening, damage by wind storms), non-anthropogenic circumstances refer to non-human induced conditions that exacerbate these disturbances (e.g., fire occurring during particularly harsh conditions like strong winds, high temperature, drought, etc.). For information on how to document that disturbances are beyond the control of and not materially influenced by the country, see Section 2.6.4 below.

The methodological guidance provided in this section is aimed at disaggregating emissions and removals in ecosystems where natural disturbances cause large IAV in emissions within the MLP and where subsequent removals occur over a multi-year period of time. Therefore, this methodological guidance is applicable to natural disturbances in forests, and in woody grassland, undrained wetlands or undrained peatlands, but not in other land categories where human actions materially determine and/or deeply influence the conditions and circumstances associated with significant emissions by disturbances (such in drained peatlands and in cropland).

Balance of emissions and subsequent removals

A fundamental assumption under the MLP is that carbon emissions and removals associated with natural effects will average out over space and time (see also Volume 4, Chapter 1). Therefore, consistent with this assumption, the CO_2 emissions (from above and below ground biomass, dead organic matter and soil carbon) from areas affected by natural disturbances are expected to be balanced by subsequent removals across the landscape at some future point in time. This expectation has no established time limit because the time to balance depends on the types of ecosystems affected by disturbances and their rates of regrowth.

At stand level, changes in growing conditions could affect this expectation, in particular if environmental conditions contribute to regeneration failure of stands that were affected by natural disturbances, e.g. landslides and erosion after wildfire, making it more difficult to achieve the balance. Conversely, if environmental changes contribute to increased growth rates or reduced mortality rates, then the balance will be achieved faster. In the case of repeated disturbances on the same area, the time to reach balance for that area may increase.

2.6.2 Relationship between different methodological approaches and the representation of emissions and removals from interannual variability

The choice of estimation method and data affects the extent to which the IAV of different drivers is reflected in reported estimates (see Table 2.6c). Countries can apply different estimation methods to report their emissions and removals capturing the anthropogenic components with different temporal resolution and disaggregation of variables (annual to periodic, averaged or disaggregated by drivers). Table 2.6c provides information on how the choice of estimation method affects whether or not factors contributing to IAV of reported emissions and removals are captured in NGHGIs. This table may help countries in understanding and describing how the various effects are reflected in the estimates of emissions and removals, therefore increasing the understanding of NGHGIs by the scientific and policy communities.

[10] Information on natural disturbance definitions and approaches applied in the Kyoto Protocol accounting can be found in IPCC. (2014) In: *2013 Revised Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol*, eds. T. Hiraishi, T. Krug, K. Tanabe, N. Srivastava, J. Baasansuren, M. Fukuda & T. G. Troxler, IPCC, Switzerland.

TABLE 2.6C (NEW)
GENERAL GUIDANCE ON WHETHER OR NOT THE ESTIMATION METHOD IS ABLE TO DISTINGUISH BETWEEN THE IMPACT OF THE INDIVIDUAL DRIVERS BELOW ON THE INTERANNUAL VARIABILITY OF REPORTED ANNUAL EMISSION AND REMOVAL ESTIMATES - NOTE THAT SOME EXCEPTIONS MAY OCCUR, DEPENDING ON THE DATA USED

			Drivers			
Method			**Direct Human**	**Indirect Human**	**Natural climate variability**	**Natural Disturbances**
Stock Difference[11] Periodic measurements (multi-year)			No	No	No	No
Stock Difference[12] Annual measurements			Yes	Yes	Yes	Yes
Gain-Loss[13]	Live biomass pools	Biomass growth based on Emission Factors or empirical yield tables	Yes	No	No	Yes
		Growth based on process (or hybrid) model	Yes	Yes	Yes	Yes
	Dead and soil organic matter pools	Dead and soil organic matter dynamics based on Emission Factors	Yes	No	No	No
		Dead and soil organic matter dynamics with constant climate	Yes	No	No	Yes
		Dead and soil organic matter dynamics with variable climate	Yes	Yes	Yes	Yes

The Stock Difference method calculates net emissions/removals (E/R) as the difference in estimated C stocks for relevant pools measured at two points in time. Average annual net E/R can be calculated by dividing the C stock difference of a period by the number of years between the two observations. Periodic stock assessments without auxiliary data therefore do not allow the quantification of the IAV of emissions and removals and its relation to the various drivers.

With annual measurements of ecosystem carbon stocks, e.g. via subsets of annual plot measurements in a continuous forest inventory, the quantification of IAV of emissions and removals becomes possible. Periodic or annual subsets of inventories can by themselves not detect IAV unless auxiliary data such as area annually burned, harvest rates or other specific plot-level measurements on the timing of tree mortality are used to inform about IAV (Röhling *et al.* 2016). For non-CO_2 emissions (e.g., CH_4 and N_2O from fires), auxiliary data on the type of disturbance that caused carbon losses would be required when the stock difference method is used.

The Gain-Loss method requires annual data on forest management, land-use change and natural disturbances and when these are available it can provide estimates of the IAV of net emissions. Depending on the estimation methodology and the data sets used, it may capture some or all of the impacts of drivers of the IAV of annual emissions and removals. A Gain-Loss approach utilising yield tables or constant emission factors (EF) will be insensitive to natural climate variability and, therefore, will only be able to distinguish between the direct human impact and natural disturbance impacts on IAV of emissions and removals. Gain-Loss methods that utilise climate-sensitive growth and mortality models (Richards & Evans 2004; Waterworth *et al.* 2007; Hember *et al.* 2018), or

[11] Forest inventories with multi-year period remeasurement and no auxiliary data cannot detect IAV. In some cases, periodic measurements on permanent sample plots are augmented with additional annual data thus increasing the ability to estimate IAV.

[12] Forest inventories with annual remeasurements for the same plots can detect IAV but are rarely implemented.

[13] The assumption for the Gain-Loss method is that activity data such as harvest, land-use change, and natural disturbances are available annually.

BOX 2.2L (NEW)

EXAMPLE OF THE TABLE FORMAT THAT COULD BE USED FOR VOLUNTARY DISAGGREGATION OF TOTAL ESTIMATED FLUXES ON MANAGED LANDS INTO ANTHROPOGENIC AND NATURAL DISTURBANCE COMPONENTS

Land-use category	**e.g. Forest land remaining forest land**					
Years		**Start year**				**Inventory year**
Total Area under the MLP (kha)						
Carbon stock change	Gains					
	Losses					
	Net					
non-CO_2 emissions	Emissions					
Net E/R plus non-CO_2	Total*					
Annual area of natural disturbances (kha)[29]						
Area subject to natural disturbances (kha)[30]						
Carbon stock change	Gains					
	Losses					
	Net					
non-CO_2 emissions	Emissions					
Net E/R plus non-CO_2	Total					
Remaining area of managed land (kha)						
Carbon stock change	Gains					
	Losses					
	Net					
non-CO_2 emissions	Emissions					
Net E/R plus non-CO_2	Total #					

* This is the total MLP estimate of net emissions and removals, i.e. the first order approximation of the anthropogenic emissions and removals

\# This is the optional disaggregated estimate of the anthropogenic emissions and removals

[29] The area of natural disturbance in the year it first occurs.

[30] The cumulative area which has been subject to natural disturbances up to and including the current inventory year, minus the area of natural disturbances on which past CO_2 emissions are considered to be balanced by subsequent removals since the occurrence of the natural disturbance. In the cumulative area totals, areas affected multiple times are included only once.